2020 International Conference on Power, Energy, Control and Transmission Systems (ICPECTS) | 978-1-7281-1084-4/20/$31.00 ©2020 IEEE | DOI: 10.1109/ICPECTS49113.2020.9336983

INDIAN AND ENGLISH LANGUAGE TO SIGN LANGUAGE TRANSLATOR- AN AUTOMATED PORTABLE TWO WAY COMMUNICATOR FOR BRIDGING NORMAL AND DEPRIVED ONES

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***Abstract*—:** Human thought are conveyed through visual manual modality. Sign languages are the kind of language that is expressed via the manual sign stream together with non-manual components. Whenever communities of deaf folks exist, sign languages have developed as handy suggests that of communication and that they kind the core of native deaf cultures. Living with a incapacity isn't simple, however the majority United Nations agency are into disability usually develop alternative heightened senses and skills, associated area unit able to live an nearly traditional life and have a heightened sense of hearing and bit, whereas the deaf and mute area unit able to browse lips and communicate mistreatment language. However most of the final population don't understand language and area unit unable to speak effectively with the deaf. Therefore the deaf realize it tough to converse with folks on daily to day basis, this issue are often solved through a smartphone application. By the application deaf and dumb folks will simply learn several languages mistreatment their convenient language severally, user will either learn in online or perhaps offline modes, additionally makes user to interactively participate and additionally provides higher level of comfort, enhances higher understanding, provides easy surroundings. The applying development is freed from value and it needs humanoid studio installation. Mobile application is absolutely adopted, can address most of the challenges facing those with hearing disorder and speech impairment.

Keywords—Sign Language, Speech recognition, speech translation , Android.

I. INTRODUCTION

Sign Languages are used to be one of the ancient languages and it can well be ahead of so many centuries as it has been appeared in the primary theories of history. It's started even before the emergence of spoken languages and this has wide selection of application. Mobile application have shown a dramatic improvement in their practicality to some extentwhereverit'scurrentlydoableto ownmobile phone execute Java programs. As a result, cellular users throughout the planetsquare measurecurrentlyable tobrowse and write email, browse websites and play java games victimization their cellular phones. This trend has promoted on propose the use of mobile application for higher communication. Signing is that the means that of communication among the deaf and mute community. Signing emerges and evolves naturally inside hearing impaired community. Signing communication involves manual and non-manual signals wherever manual signs involve fingers, hands, arms and non-manual signs involve face, head, eyes and body shown in Fig.1. Signing could be a well -structured language with a phonemics, morphology, syntax and descriptive linguistics. Signing could be a complete tongue that uses other ways of expression for communication in standard of living.

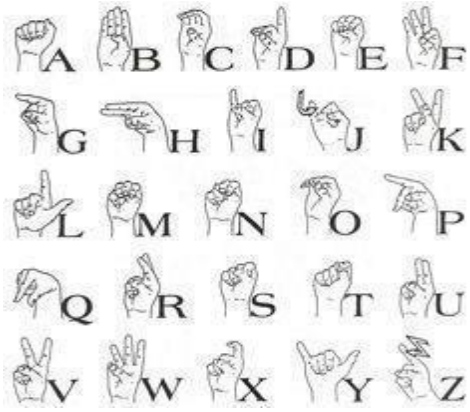


Fig.1 Sign Language symbols

Given that sign language could be a visual language, it's no surprise that you simply also can learn to higher perceive visual communication. Signing is that the communication Secretly. Whether or not its clubs, film theatres or libraries, you do not got to whisper or yell to speak to your friends in sticky state of affairs. The remainder of paper deals with the prevailing system, planned models and implementations.

These are the two different existing system works,

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Algorithms and implementations on Sign Language and are discussed below:

A) Neural Networks:-

Neural organization is furthermore called Artificial Neural Network (ANN), is a manufactured canny framework that is predicated on natural neural organization. Neural organizations fit to be prepared to play out a chose perform by changing the estimations of the associations (weight) between these parts. Neural organization is changed and prepared so as the real information winds up in a chose target yield. Model at Figure, the organization is changed, upheld a correlation of the yield and furthermore the objective till the organization yield is coordinated the objective. These days, neural organization are frequently prepared to determine a few extreme issues round looked by animal.

B) Hidden Markov Model (HMM):-

In the event of dynamic technique demonstrating, the way to deal with is chosen s random .In nature, Hidden Markov Models (HMMs) or Dynamic hypothesis Network. A period space technique represents Andre Markov property if the possibility thickness of an event, given all blessing and past occasions, relies entirely upon the utmost exceptional occasion. On the off chance that this occasion relies just upon the preeminent late past occasion, at that point the method is termed a primary request stochastic cycle. It's idea of as an accommodating presumption, though thinking about the directions of hands of a communicator for time pivot. The HMM, great for its made numerical structure, is considered to be a wide utilized device for quickly demonstrating spatial-fleeting information inside the most natural way doable.

II. PROPOSED SYSTEM

A novel system has been proposed on account of sign language translation and that will assist the deprived ones in an effective manner. This can do the real time translation from multiple Indian languages and English to sign language will help bridge this gap to large extent and can operate even without internet.

A. USE CASE DIAGRAM

Use case design for the proposed system is shown in Fig.2,which consists of usersign translator and informationas actors.ThesubsequentUseCase diagram provides high level view of Sign Translator application, additionally represents user interaction and user relationship with application.

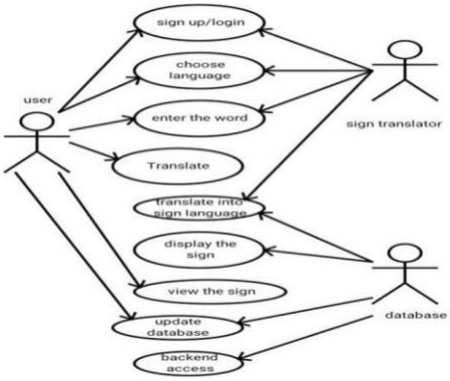


Fig.2 User interaction diagram

The user are going to be logged in to application,Sign Translator will interpret the word entered by user,which will convert it into signing exploitation information.

B. ARCHITECTURE

System architecture for the proposed system is shown in Fig.3.It consists of Mobile Unit, Sign Translator interface, Text Reader, Text Interpreter, Data Storage and Maintenance as database.

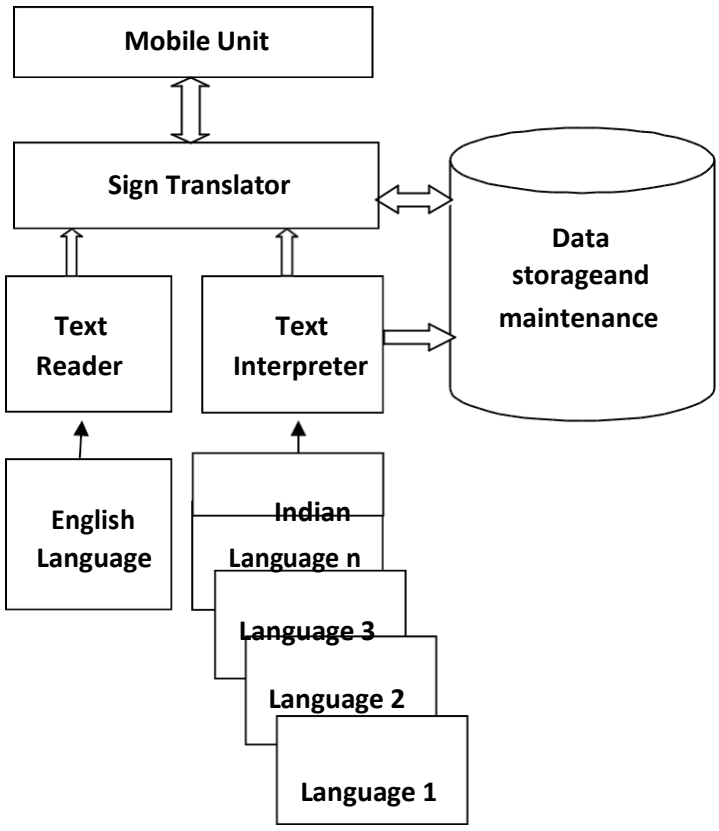


Fig.3 overall System Architecture

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The flow of the architecture is described in this section. Sign Translator will recognize speech as input, it matches the voice with string and the appropriate image associated with string will be matched and hence will be translated to Sign Language where then normal people could be able to convey their thoughts and deprived people could be able to visualize the output and can easily understand in their comfortable Sign Language. Screenshots of various components of Sign Translator results are included as evidence.

IV. IMPLEMENTATION

Deprived people are with miscommunication, Visual stimuli, and psychological situations that have been encountered in and around them. Sign translator works on the algorithms to solve these difficulties through compatible application. Some of the algorithms with appropriate pseudo code are presented in this section for better understanding of application.

Algorithm

Import necessary packages
Define class
Create objects for XML design
Set necessary permissions
Store images and word or sentences in backend
Allow the user to provide input as speech
Convert speech to string
If string matches with stored word
Then
display appropriate image

Thus above algorithm uses Speech Recognizer will get input from user, Image Viewer will display the sign language. Algorithm for speech recognizer also been presented in this paper in the following sections.

A. Speech Recognizer

Description: Speech recognizer is employed to investigate the user speech and match the speech with predefined phonetics. Speech recognizer can give recognized speech with valid word and sentence matches. The careful operating algorithmic program and design for speech Recognizer is provided below in Figure 4. Speech recognizer is employed to urge user speech in different language to convert it into signing.

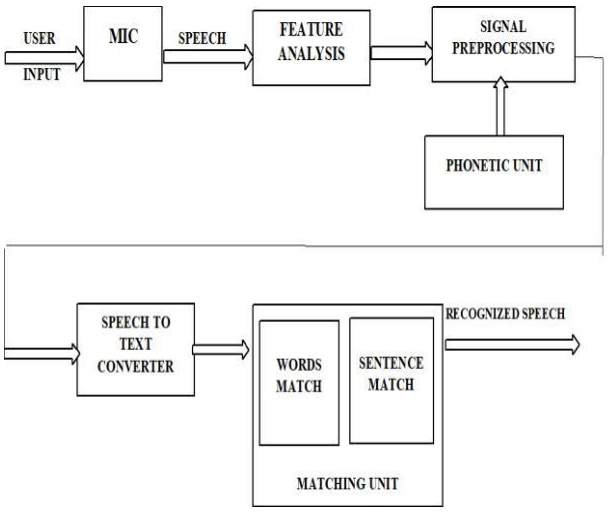


Figure 4 Design layout of speech recognizer

1) Mic:

A electro-mic transducer is associate degree instrument that converts sound to el electrical signal. Thus primarily it's a electrical device that converts one kind of energy to a different type. electro-acoustic transducer takes undulation or the variation in pressure of air as input and provides electrical signals as associate degree output. There square measure differing kinds of electro-acoustic transducer and that they use completely different techniques to convert sound to electrical signals. The foremost common electro-acoustic transducers square measure dynamic microphone, capacitor microphone, and the piezoelectric microphone. Usually microphones square measure connected to associate degree electronic equipment before it gets recorded. as a result of the electrical signals generated by a electro-acoustic transducer is extremely little and measured in mill volts. Amplifiers modify the sound quality as per the necessity. An electro-mic transducer conjointly converts sound into a tiny low electrical current. Sound waves hit a diaphragm that vibrates, moving a magnet close to a coil. In some styles, the coil moves inside a magnet. Different microphones, like condenser microphones, work on the principle of capacitance.

B. Image Viewer

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*Description :*Image viewer permits the user to envision the image as output. Viewer is that the interface for user wherever user will be ready to realize acceptable image for his or her input. Image viewer is employed to show the acceptable signs for the assorted language inputs. wherever the underprivileged may well be ready to understand in their own method through image.

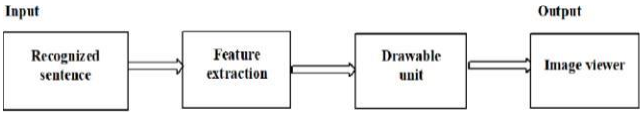


Fig.5 Image Viewer Block Diagram

Algorithm:

Recognized sentence is provided as input
MIC recognizes user speech as input
Feature analysis is done in speech
Signal processing for speech is done
Phonetic unit is added
Speech to text conversion is done
Speech forwarded for matching unit
Words match and sentence match is done
Output will be recognized speech
Feature extraction is finished
The images are going to loaded in drawable unit
Sentence with suitable image are going to be matched
The Image viewer is implemented
Output will be acceptable image for user speech

Thus, the user will read the results of their input in linguistic communication through this image viewer. Thus,it acts as interface. The image viewer will be enforced victimization predefined java package by making category object.



Fig.6 Home interface

Screen shots for the developed system is shown in the following Fig. 6.

Home interface is a main screen where user will be redirected to next page after delay to 5ms.It acts as welcome page to user.

Fig.7 shows the chat interface where user will be allowed to speak via mic and once it is recognized, it will be converted to string and string equals function will map the appropriate sign image using Image Viewer.

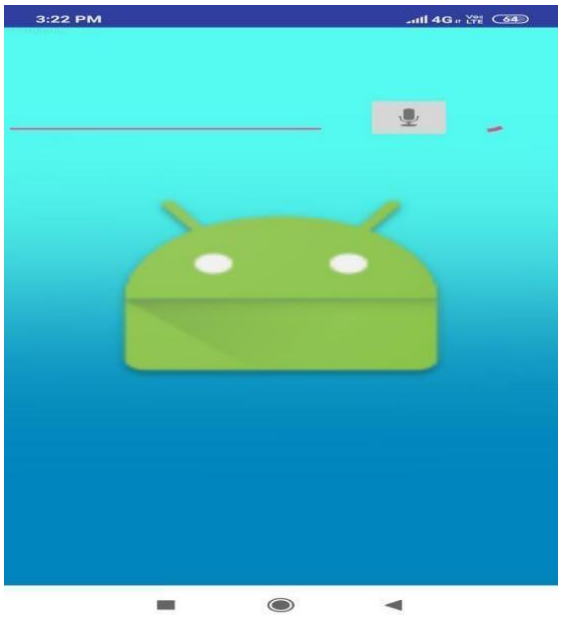


Fig.7 Chat interface

Fig.8 gives the various sample inputs and the appropriate output in the proposed system. Where some of pre-stored image is displayed for certain greetings and messages. The deprived ones can understand what the normal user trying to convey to them and thus it reduces the communication gap between them to large extents. The matching of string, image viewing and voice recognition, entire process and scenario is coded using java.XML is also used to create attractive and appropriate user interfaces using Relative layout available in Android Studio.

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Fig.8a) Hello in Sign Language

Fig.8b) Good morning in Sign Language

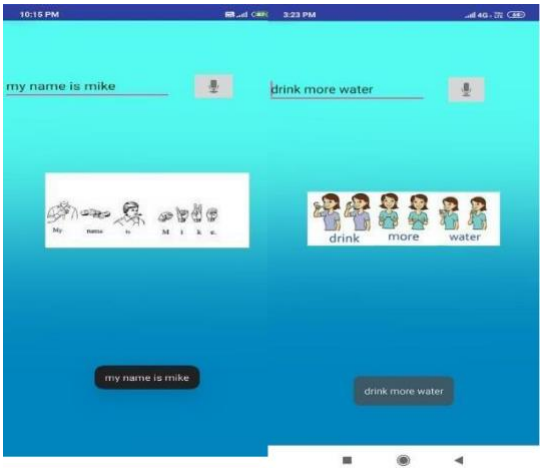


Fig.8c) Self introduction in Sign Language

Fig.8d) Drink more water in Sign Language

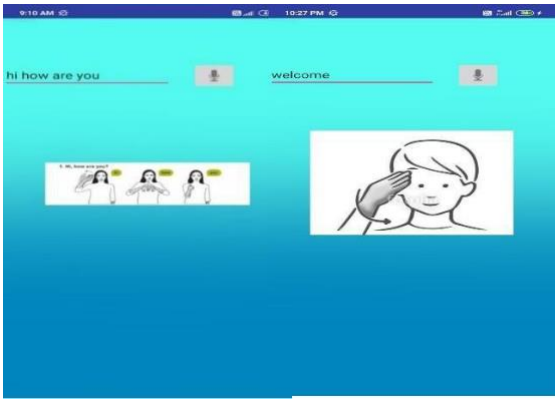


Fig.8e) Conversation in Sign Language

Fig.8f) welcome greeting in Sign Language

V. CONCLUSION

The planned system will assist the underprivileged ones throughout their interaction and created the interaction easier once they are with traditional folks. it's needed that for higher understanding of Communication through mobile, the user ought to perceive accent and sign that has been shown through the image viewer. The Mobile Application "Sign Translator" has been designed and developed in such manner, in order that the essential necessities for building these quite answer are being glad. this could be fitted with deaf folks, hearing impaired folks and traditional folks to speak with one another. Hence, there exists a large usage behind these quite solutions

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